Technical Service Bulletin

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Piston Skirt Scuffing	

Piston Skirt Scuffing

Core Issue

The purpose of this Early Field Notification is to inform the field that QST30 and QSK23,45,60,78 engines with Ferrite Cast Iron Ductile (FCD) pistons are susceptible to piston skirt scuffing, at startup, after extended periods of shutdown during very cold ambient conditions. The field study data indicates that the engine is most vulnerable when starting with oil and coolant temperatures below −12°C [10°F].

Confirmation

QST30, QSK23, QSK45, QSK60 and QSK78 engines

Some cases of high blow-by and coolant drop in the oil sump have been observed.

If an engine shows the symptoms of piston and liner scuffing, a center line inboard scuffing will be visible.

Scuffing occurs during cold starts, but may **not** exhibit any symptoms until more hours are accumulated.

The potential for piston, liner, connecting rod, crankshaft, and engine block damage exists.

Resolution

To reduce the potential for piston scuffing, it is recommended to warm the coolant and oil to a temperature of −4°C [25°F] prior to starting.

Cummins® distributors **must** work with their end user customers and/or OEM dealers to support this measure.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2020-2-26	Non-Product Problem Solving (PPS)

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